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Soundscapes in Our Hands

Designing an Interactive Tool for Non-Experts to Shape a Virtual Grocery Store Soundscape

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Dedicated to Mohammad Yaghi,

sound engineer in Gaza, committed to documenting the soundscape of the loss and destruction of Israel's genocide in Palestine.

Abstract

Everyday sound environments shape how people experience public and semi-public spaces, yet opportunities for those affected to influence them remain limited. This study investigated whether an interactive tool could support non-experts in expressing soundscape preferences, how such a tool might be designed, and what can be learned from the soundscapes participants create.

The research used a mixed-methods approach, combining field observations with soundwalks in grocery stores, sound recordings, a participatory co-design workshop, prototype development, and a user study. The prototype was a tool for controlling the soundscape of a virtual grocery store in Unity, using FMOD for audio, with a MIDI controller allowing users to adjust sound levels, apply filtering, and select between sound variations. In the user study, participants created three versions of the soundscape and reflected on their choices through questionnaires and in-depth interviews.

The results show that the tool enabled participants to express soundscape preferences and reflect on how sounds interact within an everyday environment. The design process illustrates how field listening and participatory methods can inform the development of interactive tools that are accessible to non-experts. The created soundscapes further show that perceptions of pleasantness and disturbance were shaped by context, expectations, and personal associations, and by how sounds were combined and experienced together within the soundscape.

Sammanfattning

Vardagliga ljudmiljöer påverkar hur människor upplever offentliga och semi-offentliga platser, samtidigt som möjligheterna att själv påverka hur dessa miljöer låter ofta är begränsade. Denna studie undersöker om ett interaktivt verktyg kan stödja icke-experter i att uttrycka preferenser för ljudlandskap, hur ett sådant verktyg kan utformas, samt vilka insikter de ljudlandskap som deltagare skapar kan bidra med.

Studien bygger på en kombination av metoder och kombinerade fältobservationer och soundwalks i livsmedelsbutiker, ljudinspelningar, en co-designworkshop, prototyputveckling och en användarstudie. Prototypen var ett verktyg för att styra ljudlandskapet i en virtuell livsmedelsbutik, utvecklad i Unity med FMOD för ljudhantering. Med hjälp av en MIDI-kontroller kunde användarna justera ljudnivåer, applicera filter och välja mellan olika ljudvariationer. I användarstudien skapade deltagarna tre versioner av ljudlandskapet och reflekterade kring sina val genom frågeformulär och djupintervjuer.

Resultaten visar att verktyget gjorde det möjligt för deltagarna att uttrycka sina preferenser för ljudlandskap och att reflektera över hur ljud samspelar med varandra i vardagliga ljudmiljöer. Designprocessen visar hur lyssnande i fält och participatoriska metoder kan forma utvecklingen av interaktiva verktyg som är tillgängliga för icke-experter. De ljudlandskap som deltagarna skapade visar även att upplevelser av vad som uppfattas som behagligt eller störande påverkades av kontext, förväntningar och personliga associationer, samt av hur ljuden kombinerades och upplevdes tillsammans i ljudlandskapet.

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1 Introduction

It is well established that sound can impact our health and wellbeing. Most of this research focuses on noise, typically defined as *unwanted sound* (Fink, 2019). Exposure to noise has been linked to a range of physical and mental health complications (Passchier-Vermeer & Passchier, 2000). In urban environments, noise can impair cognitive development in children (WHO, 2011). Long-term exposure can also disrupt sleep and increase the risk of cardiovascular disease (WHO, 2009; WHO, 2011).

So, *when does sound become unwanted*? Regulations from agencies such as OSHA (2025) and The Swedish Work Environment Authority (2023) typically use sound pressure level (SPL), expressed in decibels (dB SPL), to define thresholds for harmful noise. But *does sound become unwanted only when it reaches a certain level*? As Blesser and Salter (2007, p. 24) point out, the social complexity of soundscapes lies in the listener's shifting attention, which can turn something desirable into something unwanted, and vice versa. A baby's cry might be meaningful to a parent but is noise to someone nearby trying to read a book.

Furthermore, while a sound may serve an important function, it can nevertheless cause health problems. One study investigating the impact of sound on staff in preschools, maternity wards, and intensive care units found that exposure to "meaningful but disturbing" sounds, like children screaming or medical alarms, can lead to fatigue and stress, *independent of sound level* (Gyllensten, Fredriksson, Widen, & Persson Waye, 2023).

While research demonstrates that sound affects health and wellbeing in various ways, focusing solely on sound levels does not fully capture how we experience sound in everyday life or what contributes to the perception of disturbing noise. Urban planners and policymakers often rely on decibel-based noise thresholds rather than considering the broader soundscape and its subjective perception (Steele et al., 2023). Although there is extensive research on the acoustic properties of spaces, less attention has been given to how people actually experience their acoustic surroundings (Blesser & Salter, 2007, p. 11).

Researchers have sought to move beyond physical measurements toward subjective perception by studying how people experience the overall *quality* of a soundscape. A common dimension used is pleasantness (Filipan et al., 2017). However, pleasantness depends on factors such as the *type of environment*, *what people expect to hear there*, and the *cultural or historical meaning of the space* (Southworth, 1969). A lively market might be pleasant in one context but chaotic in another.

If the goal is to create pleasantness, aiming for tranquility might be more effective (Filipan et al., 2017). However, what is perceived as tranquil is subjective and personal. One study by Marry and Defrance (2013), combining acoustic measurements with surveys and interviews

in Parisian public squares, showed that participants' perceptions often conflicted with acoustic data. One location with *high sound levels* from fountains and street activity was described as *pleasant and lively*, while some *quieter locations* were seen as *boring* or *unsettling*. It therefore appears that our expectations of an environment influence how we experience its soundscape. Reactions to sound are also shaped by personal memories and cultural associations (Blesser & Salter, 2007, p. 13).

To understand how soundscapes are experienced, it seems necessary to ask those who experience them. However, this approach also has challenges. Most people *lack a precise, widely understood language for describing auditory experiences*, in part due to a culture that prioritizes the visual and undervalues the emotional importance of hearing (Blesser & Salter, 2007, p. 6). Even when people have the language to talk about sound, *they may not reflect on it in daily life or be aware of which soundscapes they enjoy or dislike* (Blesser & Salter, 2007, p. 8). The mental process of making sense of sound is complex and still not fully understood (Blesser & Salter, 2007, p. 12).

The problem appears to be twofold. First, *there is a lack of participatory input from those who actually experience the soundscape*. Designers and managers of these environments often overlook the subjective impact of sound on those who inhabit them. Second, even when such impacts are considered, *existing methods for assessing soundscape quality do not fully capture how sound is perceived or lived*. The issue is also political, as sound is tied to access to power and functions as a tool of power. Those who own or manage a space determine what is heard while others have limited ability to control their exposure to different sounds (Blesser & Salter, 2007, p. 33).

This raises a question: *how can we understand the preferences of those who experience the soundscape if there is little shared ground for expressing them*? Making this process more accessible could support different forms of influence. It could help designers take others' needs into account or allow groups, such as unions in workplaces, to advocate for concrete changes to the sound environment.

1.1 Research Questions

The aim of this thesis is to explore whether an interactive tool can support people who do not work professionally with sound, who I refer to as non-experts, in expressing their preferences in everyday soundscapes. The research is guided by the following questions:

RQ1*: Can an interactive tool support non-experts in expressing their soundscape preferences and reflecting on their everyday sound environments, and if so, how?*

RQ2: *What considerations and methods inform the design process of such a tool?*

RQ3: *What insights can be gained from the soundscapes created by participants using the tool?*

1.2 Scope of Study

This thesis focuses on the soundscapes of grocery stores, which I argue are a compelling site for investigating the research questions. These spaces are privately owned but, unlike other retail environments, most people need to visit them regularly for food and essentials. In this sense, they function as a kind of semi-public space. Unlike outdoor environments such as parks, grocery stores are enclosed, making their soundscapes easier to simulate.

Participants in this study were between 20 and 40 years old. This reflects the practical constraints of a master's thesis rather than a specific intention to target this age group.

1.3 Thesis Structure

The thesis is structured as follows. Section 2 provides background on soundscapes and relevant concepts that inform the study. Section 3 describes the methodology and design process, including field observations, a co-design workshop, prototype development, and user study. Section 4 presents the results of the user study. Section 5 discusses these findings in relation to the research questions, addresses limitations, and outlines future work. Section 6 concludes the thesis.

1.4 Ethical Considerations

This study follows the principles of good research practice outlined by the Swedish Research Council (Vetenskapsrådet, 2024), emphasizing reliability, honesty, respect, and accountability. Participation was voluntary, with informed consent obtained after participants were informed about the study's purpose, procedures, and their right to withdraw at any time.

No sensitive personal data was collected. Only age and previous experience with working with audio were gathered. Parts of the interviews were recorded for accurate transcription and stored securely with access limited to the researcher. Recordings and questionnaire data were anonymized during transcription to protect participant confidentiality.

Data handling followed GDPR and ethical guidelines, ensuring participants' privacy and integrity. The study aimed to contribute to knowledge supporting healthier and more participatory sound environments while minimizing potential risks to participants.

2 Background

Understanding how soundscapes are defined, studied, and experienced is necessary for situating this thesis within existing research. This section outlines the theory and research that inform this work. It presents key concepts from acoustic ecology and soundscape studies, examines how soundscapes are experienced in everyday environments—specifically grocery stores—and reviews different approaches to categorizing sounds. Lastly, it presents research on interactive soundscapes.

2.1 Soundscapes and Acoustic Ecology

The term soundscape was initially defined as "*an environment of sound (sonic environment) with emphasis on the way it is perceived and understood by the individual, or by a society*" (Truax, 1978, p. 126). A standardized definition proposed in 2014 describes it as "*the acoustic environment as perceived or experienced and/or understood by a person or people, in context*" (International Organization for Standardization, 2014). Soundscape Ecology, or Acoustic Ecology, is an interdisciplinary field combining landscape ecology and acoustic studies to *examine the relationship between living beings and the soundscapes they experience* (Droumeva & Jordan, 2019).

The field was started by the World Soundscape Project (WSP), founded by composer, writer and environmentalist R. Murray Schafer in the late 1960s in Vancouver, Canada (Truax & Barrett, 2011). The project emerged from a growing concern that noise pollution was transforming Vancouver's soundscape from a *hi-fi* environment, where individual sounds were acoustically clear and spatially distinct, into *a lo-fi* one filled with constant, overlapping noise that masked subtle auditory details (Truax & Barrett, 2011). Schafer and his colleagues sought to document and protect sonic environments characterized by acoustic clarity and balance, often associated with rural or pre-industrial settings, which they viewed as increasingly threatened by urban development and technological expansion (Truax & Barrett, 2011).

This focus on preserving hi-fi soundscapes has received criticism from scholars for romanticizing western pre-industrial environments and having limited applicability to complex urban realities (Droumeva & Jordan, 2019; Akiyama, 2019). Moreover, notions of a "healthy" soundscape have often reflected class and gender hierarchies, with silence seen as a marker of class in contrast to the sounds of working-class environments (Bijsterveld & Krebs, 2013).

A different framework for understanding soundscapes is the concept of *ecotonality*, coined by acoustic ecologist, composer, and sound artist Andra McCartney (2016). McCartney argues that overlapping sounds, often dismissed as noise in the lo-fi/hi-fi framework, are not necessarily "bad" or ecologically unsound. Drawing on the ecological concept of

ecotones, which are transitional zones, where ecosystems meet and interact, McCartney uses ecotonality to describe the layered and shifting nature of sound environments, particularly in urban and post-industrial contexts (Droumeva and Jordan, 2019). It refers to complex environments where different sonic elements coexist, overlap and interact.

2.2 Experiencing Soundscapes in Grocery Stores

Building on the concept of ecotonality, the soundscapes of public and semi-public spaces can be understood *as layered, shifting environments shaped by overlapping and interacting sound sources*. These environments differ from those typically studied in traditional auditory research, which has often focused on clean, tonal sounds under controlled conditions, such as in musical or laboratory settings (Gaver, 1993a).

Grocery store soundscapes are a good example of these layered environments. When you think of the sounds of a grocery store, you'll find a wide range of clips: the *ka-ching* of the cash register, the *beeps* of the scanner, the *rolling* sound of a shopping cart being pushed, the *humming* of freezers, and the *voices* of customers and staff. The soundscape is layered, constantly shifting, and often includes sounds that have not been designed with the overall soundscape in mind.

For example, notification and alarm sounds in stores are often intended to capture attention but can end up feeling disruptive or intrusive. One study by Arfvidsson, Eriksson, Lidbo, and Falkenberg (2021) showed that such sounds can still be effective if they are lower in volume than background music, short, and not reliant on a sharp attack to be noticeable. Sounds that did not blend into the soundscape were not necessarily more effective than those that did.

In retail contexts, the research has predominantly focused on using soundscapes to influence customer behavior (Helmefalk & Hultén, 2017; Biswas, Lund, & Szocs, 2019; Jung, Kahl, & Spassova, 2011). One such example is a case study conducted in a large Finnish supermarket, which examined whether nature soundscapes in the produce section could impact sales figures (Kontukoski, 2018). The results were mixed and somewhat unclear regarding direct sales effects. However, the nature soundscape seemed *to encourage customers to pause, reflect on their surroundings, and engage in social interactions*. Participants also reported distinct preferences, *finding continuous forest sounds calming* and effective at making the space feel larger, while *sporadic sounds, such as bird calls, felt unnatural or out of place*.

We do not yet fully understand how soundscapes influence people's behavior in public places, for example, whether certain sounds make people feel safe, relaxed, or uncomfortable, and how people's actions, in turn, shape the sound environment (Aletta & Xiao, 2018). Aletta et al. (2016) highlight a disconnect between behavioral response and

subjective evaluation. In their study, participants stayed longer in a space when music was present, but this did not result in more positive ratings of the environment.

Since the present study does not focus on customers as a consumer group, but rather on people who experience the soundscape most directly, it is relevant to consider research on how employees in these environments perceive and are affected by the soundscape. Employees, which includes shop-floor staff, cleaners, and security personnel, arguably experience the soundscape to its fullest, primarily because they spend extended periods in these spaces. Furthermore, being present through hearing and listening can be part of their work tasks, such as communicating with customers and colleagues.

In contrast, customers can, for example, create a private sound environment by using headphones (Barns, 2019). This allows them to block out unwanted sounds but also reduces their awareness of the surroundings and may limit opportunities for social interaction. Workers, however, are typically not able to filter or escape the soundscape in the same way, and are therefore more continuously exposed to it throughout their shift.

However, there is limited research on how retail workers perceive the soundscapes of their workplaces. The only example I found focused specifically on music. A large-scale survey of Swedish retail employees examined how in-store music influenced their work environment (Carlén, 2018). The study found that music could be energizing, stressful, or distracting, depending on what was played and *whether staff had any influence over it*. Positive responses were more common when workers had some control over the music or when playlists were varied, while repetitive music without staff input was described as tiring or irritating. In stores without music, 71 percent said they would prefer to rather have music to improve motivation. These findings suggest that *control over the sound environment shapes how it is experienced*, and that *music is often seen as something that can improve the work environment*.

If we view the soundscapes of grocery stores through the lens of semi-public spaces, additional research becomes relevant. One study of how people experience soundscapes in public and semi-public places focused on the role of the listeners expectations (Bruce & Davies, 2014). The researchers found that *people's expectations significantly influenced their emotional responses to different soundscapes*. For instance, participants were less annoyed by unpleasant sounds when these were anticipated, whereas unexpected sounds breaking established norms caused greater annoyance. Additionally, the expected activity within a space influenced their reaction to its soundscape. In areas which people usually just pass through, like in a busy street, the soundscape had minimal impact on annoyance levels, as sounds typically matched expectations. Conversely, *participants experienced frustrations when the soundscape hindered expected tasks*, such as hearing a phone ring or understanding station announcements clearly (Bruce & Davies, 2014).

2.3 Taxonomy of Sounds

When researching soundscapes, there is often a need to categorize and group different sounds. There are multiple ways to categorize sounds, depending on the focus of the classification system. Schafer's (1977) typology of sounds focuses on their perceptual and cultural roles within a soundscape. In his framework, environmental sounds are categorized into three main types. *Keynote sound*s form the ambient background and are often heard unconsciously, such as traffic or ventilation. *Sound signals* are those that are consciously attended to and serve a communicative function, such as announcements at a train station or a ringing bicycle bell. Lastly, *soundmarks* are distinctive sounds specific to a location that hold cultural or social significance, such as the call to prayer in a neighborhood mosque or the distinctive melody of an ice cream truck. Another way to categorize sounds is by their source. An example of this is within urban sound research, with the Urban Sound Taxonomy (Salamon, Jacoby, & Bello, 2014). The taxonomy was developed by combining earlier sound classification systems and by analyzing over 370,000 noise complaints submitted through New York City's public service hotline. It categorizes sounds into four top-level groups: human, nature, mechanical, and music.

These examples show that different systems highlight different aspects, and the way sounds are grouped often depends on the context in which the classification is used, and what the researcher is trying to understand.

2.4 Interactive Soundscapes

Research on interactive soundscapes and on tools that allow individuals to influence their auditory environments highlights how participatory and responsive sound technologies can reduce the negative effects of noise for those most exposed to harmful or disturbing soundscapes. Although this field is still limited, two studies are particularly relevant for this investigation.

Ding, Meng, Yang, Zhao, & Berardis's study (2024) explored whether an interactive soundscape tool could influence how people experienced comfort in a simulated urban park in Harbin, China. A base layer of traffic noise was used to create a noisy park environment. Participants could select from twelve types of sound insulation, such as shrubs, dense trees, or soundproof walls. These options adjusted the level of traffic noise to simulate how different materials might reduce sound. Participants could also add other sounds, including birdsong, rain, ocean waves, or construction noise. After making changes, they were asked to rate how comfortable the environment felt. Preferences shifted with noise level. Natural sounds were generally preferred, and more changes were made when the traffic noise was louder or more stressful.

The study that most closely resembles the aim of this thesis used an interactive soundscape simulator to support reflection among urban professionals, such as planners, designers, and

architects (Tarlao, Steele, Blanc, & Guastavino, 2023). The study was set up in a music hall with 48 speakers arranged in an ambisonics configuration. Participants were to pick and choose between over twenty recorded sounds, including human activity, natural elements, mechanical or urban sounds, and artistic or cultural sounds. These could be placed across a spatial map of a public square, with some following predefined trajectories, such as a bus moving through the space. Participants did not control the interface directly, but instructed two operators to, for example, “add bird sounds near the trees” or “remove the ventilation unit”. The participants were asked to create three types of soundscapes: one ideal, one unpleasant, and one realistic version that they could modify. The tool was described as helpful in supporting reflection and encouraging new ways of thinking about sound. The participants also noted the importance of including perspectives beyond those of experts, and pointed to the value of involving people who use or inhabit the space being designed.

3 Methodology and Research Framework

This thesis uses a mixed methods approach, combining qualitative and quantitative methods (Bryman, 2006). This choice aligns with justifications proposed by Greene et al. (1989) and expanded upon by Niglas (2004). It follows a complementarity logic, where different methods help clarify and expand each other's results. It also reflects a developmental approach, where one method informed the next, and an explanatory one, where one method helps make sense of the results from another. Lastly, it supported the goal to build a more complete understanding by drawing on different types of data.

The work also follows a *Research through Design* (RtD) approach, where design is treated as a mode of inquiry and a way of generating knowledge (Zimmerman, Forlizzi, & Evenson, 2007). RtD positions the designer as both researcher and practitioner, recognizing that their judgments, values, and experiences shape the research process and acknowledging that knowledge generated is context dependent.

This thesis uses the International Organization for Standardization (2014) definition of soundscape as "*the acoustic environment as perceived or experienced and/or understood by a person or people, in context*". This definition shapes the framework by acknowledging that listening is not a neutral act but one shaped by its contexts. In this thesis, this has motivated an emphasis on methods that engage directly with how sound is lived and felt, while also recognizing that such engagement can take different forms depending on the setting and purpose of the study.

Lastly, the research draws on *participatory design methods*, which originated in 1970s Scandinavia through union-driven projects where workers participated in designing computer systems they would use on the job (McShane & Gustafsson, 2024). In my study, participants were treated not as subjects but as co-creators, whose reflections, preferences, frustrations and satisfactions shaped both the design of the prototype and its evaluation.

3.1 Overview

The research process followed an agile developmental structure, with each method building on the outcomes of the previous one. I began with *field observations* to understand the soundscapes I was working with. These informed the selection of a store for *sound recording*, which was used to produce a curated soundwalk, a recorded simulation of a typical shopping journey. This soundwalk was then used in a *co-design workshop* to prompt participant reflections on everyday soundscapes and ideas for how the tool could be designed. Findings from the fieldwork and workshop shaped the design of an *interactive prototype* that allowed users to create and adjust the soundscape of a virtual grocery store. The prototype was evaluated through a *user study*, where participants created different

soundscape versions and reflected on their auditory experience through a questionnaire and in-depth interview. An overview of this process can be seen in Figure 1.



Figure 1: Overview of the research methodology and design process

3.2 Field Observations

3.2.1 Aims and Rationale

The aim of the field observations was to gain a grounded understanding of the grocery store environment and its soundscape, including spatial layout, acoustic characteristics, and how it affected my experience as a listener.

To do this, I used soundwalking, defined by Westerkamp (1974) as "*any excursion whose main purpose is listening to the environment*". While Westerkamp questioned its effectiveness in urban settings, later research supports its use in these contexts (Adams et al., 2008; Jeon, Hong, and Lee, 2013; O'Keeffe, 2015). Gaver (1993a, 1993b) further connects listening to movement, describing sound as an auditory array that shifts with position, making movement necessary to perceive the full soundscape.

I also used an autoethnographic approach to notetaking, meaning the notes were written from my own perspective as a listener (Adams, Holman Jones, & Ellis, 2014), as it aligned with the overall research framework. The notes aimed to produce thick, qualitative descriptions to inform which sounds to include in the prototype and to support selecting a store that represented an "average" acoustic environment for recording.

3.2.2 Procedure

Fieldwork was conducted between 17 and 20 March 2025 in seven grocery stores across Stockholm. The stores varied in brand, size, layout, and location. Small to medium-sized stores were prioritized, as I did not want differences in soundscape to depend mainly on size.

In each location, I walked slowly through the store from entrance to exit, stopping to take notes written as a stream of consciousness. As a guide, I had prepared prompts and questions inspired by Westerkamp's (1974) *Soundwalking*, including:

- *What sounds are taking your attention? What sounds can you not hear unless you focus?*
- *How do they make you feel?*

- *What, if any, are the reactions of the people to these sounds? Does it impact people's behavior in any way?*
- *What is the sound with the lowest pitch? What is the sound with the highest pitch?*
- *Are there any rhythmic sounds?*
- *Human sounds? Mechanical sounds? Can you hear nature anywhere?*
- *How does the sound change, from this location to the other?*

During the soundwalks, I shifted between different modes of listening. At times, I focused on the source and meaning of sounds, which Gaver (1993a) describes as everyday listening, where attention is on what is happening, where, and why. At other times, I focused on qualities like pitch, rhythm, and texture, which Gaver (1993a) refers to as musical listening. I also alternated between attentive listening, focusing on individual sound sources, and holistic listening, taking in the full soundscape without isolating elements (Botteldooren et al., 2016).

Occasionally, when store personnel were curious about my presence, I spoke with them informally about their thoughts on the store's soundscape. These conversations were not part of the formal data collection but helped me, as a designer, better understand their experiences.

3.2.3 Data Analysis

The notes were coded (Lazar et al., 2017, p. 303) and analyzed, focusing on descriptions of the sounds, their locations within the store, and how they affected my experience as a listener.

3.2.4 Outcomes

This section presents the outcomes of the field observations.

Each soundwalk took 20 to 45 minutes, depending on the store's size and the complexity of its soundscape. Across all seven stores, recurring sounds formed the backbone of the grocery store soundscape. I grouped these into categories inspired by the source-based Urban Sound Taxonomy (Salamon, Jacoby, & Bello, 2014), adapted to the grocery store context:

- **Mechanical and appliance sounds:** Continuous hums from refrigerators, freezers, and ventilation, often resembling white or brown noise and carrying across zones. One freezer section was described as "sounding like a server room".
- **Customer-generated sounds:** Bag rustling, footsteps, ambient conversation, and mobile phone notifications were consistently present. Paper and plastic bags produced varied textures, sometimes sharp and crackling, other times soft and ambient, like fire.

- **Staff activity:** Sounds from stocking, box-cutting, and rolling carts were distinct and often louder than customer activity. Staff carts were described as “more confident and forceful”.
- **Transactional sounds:** Checkout beeps, receipt printers, and self-checkout interfaces. At some checkouts, scanning beeps came in bursts. At others, they felt rhythmic.
- **Environmental leakage:** Passing cars, wind, and motorcycles filtering in through entrances.
- **Music:** Background music was present in some stores, varying in volume and quality. Abrupt transitions between music zones created inconsistencies in the auditory experience.

I also noted how the soundscape changed across different parts of the store:

- **Entrance areas** were perceived as chaotic, with squealing automatic doors, street noise leakage, and shifting foot traffic. Checkout sounds were often heard loudly.
- **Produce sections** were softer, with bag rustling and quiet footsteps dominating. In one store, these textures were likened to a “market-like atmosphere”.
- **Refrigerator areas** featured dense mechanical and tonal hums that filled the space with a “steady drone sound”.
- **Freezer sections** felt very immersive. While not necessarily louder, the sound was described as “overwhelming and claustrophobic”, creating a sense of being enclosed by sound.
- **Dry goods aisles** were quiet, with minimal background noise, allowing small, transient sounds like a single voice or bag rustle to stand out.
- **Checkout areas** were the most complex, with beeping, conveyor belts, footsteps, voices, and bag noises.

Sometimes these zones transitioned abruptly, such as moving from the freezer section to the dry goods aisle, while other times they blended together, as with the entrance and checkout areas.

Furthermore, certain sounds consistently stood out as attention-demanding or unpleasant:

- **Cartwheels** with high-pitched resonance on tiled floors, described as grating or “dragging”.
- **Metallic collisions** from carts clashing or shelving being moved.
- **Alarms and alerts**, such as entrance beeps.
- **Phone sounds**, including speakerphone conversations, ringtones, and alerts.
- **Screaming and crying children**.

Sometimes the listening experience itself produced interesting outcomes. In one store, a high density of unpleasant sounds led to fatigue and irritation, prompting a shift from attentive to holistic listening. In quieter stores, the dominance of mechanical noise and absence of human sounds felt “dystopian”. Stores without music often felt exposed or sterile, while those with uneven sound systems produced a disjointed soundscape.

These field observations guided the choice of store for the next step of sound recording but were mainly used to define the key sound groups and spatial layout for the prototype.

3.3 Sound Recording and Editing

3.3.1 Aims and Rationale

The aim of this phase was to create a curated “soundwalk” for participants to listen to during the co-design workshop. The soundwalk was intended to document a representative auditory journey through a typical grocery shopping experience. Since organizing actual soundwalks in stores was not practical, a recorded and edited version was a more suitable alternative. As Gallagher and Prior (2013) describe, sound recordings allow listeners to “*suspend disbelief – to imagine that they are in the place where the recording was made*” (p. 275). The recording aimed to support this imaginative engagement by helping participants focus on how the store might feel and sound if they were there.

I selected the store for recording based on its complex soundscape, which included all the recurring sounds and zones identified during the field observations.

3.3.2 Procedure

To record the grocery store soundscape, I used a Zoom H3-VR microphone, which captures ambisonic (360°) audio. I walked through the store at a slow pace, resulting in a seven-minute recording. The recording was edited in Reaper (Cockos Incorporated, 2025), which supports ambisonic audio, and cut down to a four-minute sequence suitable for the workshop while remaining representative of the soundscape and zones (Figure 2). Crossfades were used between cuts for smooth transitions, and identifiable conversations were removed to protect privacy.

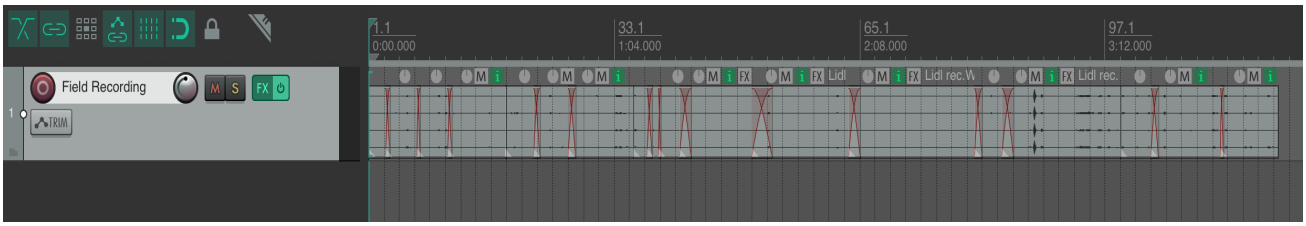


Figure 2: Screenshot of the edited ambisonic grocery store recording in Reaper.

3.4 Co Design workshop

3.4.1 Aims and rationale

The workshop had three aims: to *explore how participants reflect on everyday soundscapes*, to *gather their reactions to the curated grocery store soundwalk*, and to *collect their design ideas for the tool and thoughts on alternative auditory futures*.

In participatory design, it is recommended to involve users early in the process (Xiao, Lavia, and Kang, 2017). For this workshop, I included a mix of participants with and without sound experience. Since the tool is intended for non-experts, their involvement was important, while input from those with audio expertise could prompt reflection among non-experts and support discussion on how sound is experienced and interpreted.

A common challenge when working with non-experts is that they often find it difficult to describe how sounds make them feel, lacking the language to articulate their experience (Blesser & Salter, 2007, p. 13). To address this, I included sketching as part of the workshop, both to help participants express their thoughts and to support reflection and conversation (Marentakis, 2024). The workshop results complemented the field observations and directly informed the prototype's features.

3.4.2 Procedure

The workshop took place on April 13, 2025, in a quiet meeting room and lasted about two hours. Participants were recruited through social media and word of mouth. All gave informed consent. Partial audio recordings were made for transcription and deleted afterward, and I also took notes during the group discussion.

The workshop was structured into three parts:

1. **Icebreaker and Listening Awareness**
   This part aimed to ease participants into reflecting on and discussing soundscapes as a subjective experience. Prompts encouraged them to consider sounds they find calming, irritating, or stressful, as well as their thoughts on transitions between different everyday soundscapes.
2. **Individual Listening and Sketching**
   Participants listened individually to the curated soundwalk using Sony WH-1000XM4 headphones, played directly through Reaper. While one participant listened, the others sketched their reflections on blank canvases. The prompts were written on a whiteboard:

- what sounds they noticed or disliked
- what sounds they might want to change, remove, or amplify,
- and what kind of interaction or tool might enable that.

3. **Group Discussion**

This part aimed to gather collective reflections and design ideas. Participants presented their sketches, followed by an open discussion of their thoughts on the soundscape and possible tool designs.

3.4.3 Data Analysis

The group discussions were transcribed, coded, and analyzed chronologically alongside my notes. The canvases were photographed and used to complement the analysis.

3.4.4 Outcomes

Five participants (two non-binary, one man, two women) aged 23 to 30 (M = 27, SD = 2.6) took part in the workshop, including two with professional sound experience and three with no audio background.

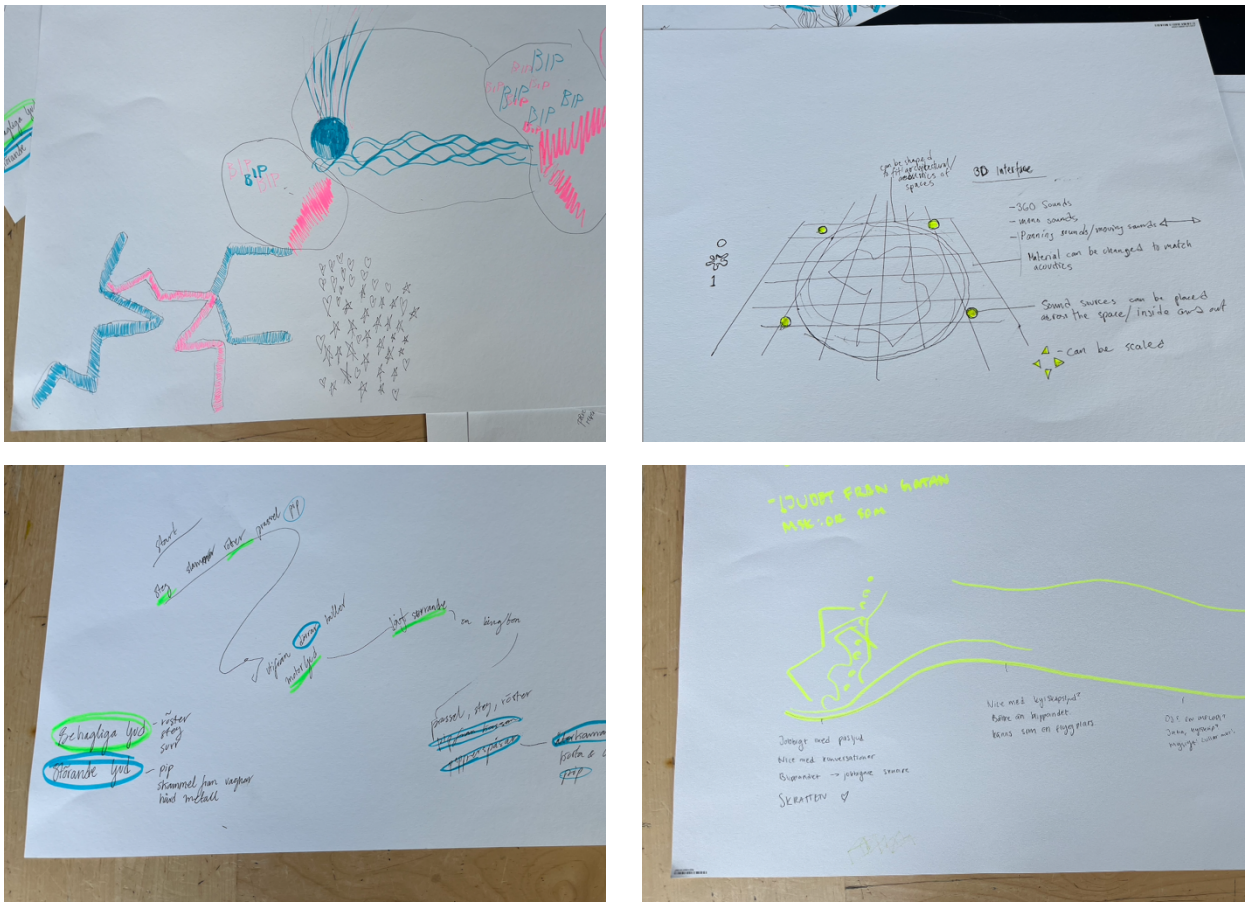
In the first part, participants shared a preference for *soft, continuous*, and *natural* sounds. They described brown noise, refrigerator hums, ocean waves, wind in trees, and birdsong as calming. They also identified sounds they found unpleasant, overstimulating, or irritating, including *high-frequency sounds* like *subway screeches*, *speakerphone conversations*, and the *sharp rustle of plastic or paper bags*.

Participants expressed discomfort in environments like cafés and public transport, noting that r*everberation*, *close proximity*, and *competing sensory cues* felt overwhelming. One described these settings as a "*witch's brew of sounds*". They also discussed transitions between everyday soundscapes, such as moving from a noisy street into a quiet library or from an echoing stairwell into a carpeted hallway.

During the group discussion, participants presented their canvases (seen in Figures 3–6). Some focused on reactions to the soundwalk, while others developed design ideas for the tool. Reactions to the soundwalk varied by section.

The entrance and checkout areas were described as *stressful and cognitively exhausting*, with *checkout beeps drawing strong reactions*. Several questioned why these beeps needed to cut through the soundscape, leading to a discussion of their social implications. One participant described them as signals of "*mutual surveillance*", suggesting they served control rather than function. Participants agreed checkout beeps should be redesigned if needed for feedback or removed altogether. *Paper bag sounds* were also seen as *uncomfortably sharp*, described as "*cutting into the ears*".

In contrast, refrigerator and freezer sections were seen as *calm, immersive, and comforting*, compared by participants to a spaceship, a gentle choir, or an enveloping field. Distant conversations and laughter were also received positively, described as a "*breath of fresh air*". Participants expressed a desire to prolong time in pleasant areas while avoiding or shortening time in stressful parts.



Figures 3-6: Pictures of some of the participants' canvases showing identified pleasant and unpleasant sounds, reactions to the soundwalk, and sketches of possible sound design features.

Participants also shared their ideas for the tool's design. A recurring theme was the *wish to reduce or suppress stressful sounds*, while *amplifying sounds they liked*.

Several participants proposed visual interfaces, including a *floating interface that could move with the user*. Other suggestions included *spatial placement of sound sources*, *directional controls*, and *interfaces for trimming*, *previewing*, *or isolating sounds in real time*. One participant envisioned a walkable virtual space where sounds could be adjusted through *embodied gestures*. Participants also reflected on how auditory design shapes social behavior and well-being. Checkout beeps and alarms were discussed as *tools of control and monitoring*. They discussed *redesigning soundscapes to feel more humane, supporting autonomy, calm, and a sense of privacy in shared spaces*. They also discussed using *different materials* in stores to influence the acoustic environment.

Some differences emerged between expert and non-expert participants. Non-experts described sounds by *source* or *reaction on them*, using phrases like "*that sound was annoying*" or "*it sounded like an alien spaceship*". Experts used these descriptors but added technical terms like "*high frequency*", "*ambient sounds*", or "*soundscape mix*". Their design

ideas often resembled a 3D digital audio workstation with detailed visual interfaces and spatial controls. In contrast, non-experts focused on which sounds to reduce or amplify, and gave broader suggestions for the grocery store soundscape.

The workshop identified which sounds participants found desirable or stressful and gathered their ideas for adjusting a grocery store soundscape. These insights informed the sounds and interaction features used in the prototype.

3.5 Prototype Development

3.5.1 Aims and rationale

Based on findings from the fieldwork and workshop, I developed an interactive tool for controlling the soundscape in a virtual grocery store. The prototype was designed to support non-experts in creating, adjusting, and reflecting on the soundscapes of grocery stores in particular, but also on soundscapes more generally.

3.5.2 Procedure

When developing the prototype, I focused on enabling intuitive interaction with the virtual grocery store soundscape. An earlier design explored an in-game drag-and-drop interface for placing and moving sound sources, based on workshop ideas, but this was abandoned for being too visually focused and complex for non-experts. To still incorporate embodied control, I used an external Korg nanoKONTROL2 MIDI controller (Figure 7), which has eight programmable sliders, each with a fader, three buttons, and a rotary knob.



Figure 7: The Korg nanoKONTROL2 MIDI controller

The visual environment was built using Unity, with assets from a commercial modern grocery store package (AndragorInc, 2021). Its visual layout (see Figure 8 and 9) was based on typical store spatial layouts observed during fieldwork.



Figure 8: Screenshot from Unity showing the interior of the virtual grocery store environment used in the prototype.



Figure 9: The store layout from above. The spatialized sound sources are circled in red.

To let users walk around the store, I implemented a conventional first-person system using WASD keys for movement and the mouse for looking around. Audio was handled with FMOD Studio (Firelight Technologies, 2025a) integrated into Unity via plugin (Firelight Technologies, 2025b), as Unity's native engine lacked robust spatial audio and real-time control. The MidiJack plugin (Takahashi, 2020) connected the controller to Unity.

I wanted the tool to be customizable enough for users to express their soundscape ideas while remaining accessible to non-experts. I therefore assigned the eight channels to different sound groups based on field observations, mapping three variations to the three buttons on each channel (Table 1). The A, B, and C options did not follow any shared pattern across channels but represented distinct versions chosen for each sound group individually.

Table 1: Overview of sound groups, spatialization, and variations assigned to each channel. A, B, and C correspond to the S, M, and R buttons on the NanoKONTROL2 controller, respectively.

Channel	**Sound Group**	**Spatialization**	**Variations**
1	*Bags*	*Produce section*	A. *Soft plastic* B. *Loud paper* C. *Muffled impact*
2	*Carts*	*Entrance and checkout*	A. *Light hollow roll* B. *Metallic cart* C. *Loud clunky carts*
3	*Checkout*	*The checkout area*	A. *Soft digital beeps* B. *Checkout beeps* C. *Conveyor belt only*
4	*Freezer*	*The freezer area*	A. *Subtle hum* B. *Airy hollow tone* C. *Intense tonal hum*
5	*Music*	*Not spatialized*	A. *Very calm ambient* B. *Chill piano* C. *Rhythmic club music*
6	*Footsteps*	*Not spatialized*	A. *On carpet* B. *On wood* C. *On concrete*
7	*Room Tone*	*Not spatialized*	A. *Small indoor area* B. *Outdoor breeze* C. *Industrial ventilation*

8	*Voices*	*Not spatialized*	A. *Airport gate talk* B. *Phone call* C. *Youth chatter*

The variations were based on field observations and workshop insights. I noted differences in *bag sounds*, such as paper versus plastic, which also came up in the workshop. *Cart sounds* also varied in intensity between staff and customers. Discussions about *checkout beeps* led to three versions: a soft digital beep for cashier feedback, a loud scanning version, and one with only the conveyor belt. For *freezers*, I created versions ranging from low-frequency hums to more tonal sounds based on field observations. The *music* variations included calm ambient, upbeat piano, and rhythmic electronic to provide range in genre. *Footstep* sounds reflected different floor materials, inspired by discussions on how materials affect acoustics. *Room tone* was included to prevent empty-sounding spaces, using recordings from a living room, an outdoor setting reflecting preferences for natural sounds, and a vent-heavy warehouse. *Voice recordings* were the most challenging to source. I used muffled airport voices, a loud phone call reflecting participant complaints and my own observations, and a recording of teenage chatter.

To make the sounds, I used foley recordings from the commercial library Splice (Splice, 2023), with additional editing in Logic Pro X (Apple Inc., 2023). Each variation was made as a loop, ranging from 48 seconds to just over one minute. I used crossfades and volume automation to make the loops feel realistic and balanced their loudness with a loudness meter. Most categories were mixed between −20 and −15 LUFS Loudness Units Full Scale), while room tone was mixed around −30 LUFS to maintain a background presence. An adaptive limiter with a ceiling of −1 dB was applied to prevent spikes. All loops were exported as WAV files at 48,000 Hz.

Each of the eight sound groups was implemented as an FMOD event with a parameter set to 0, 1, or 2 to activate one of three variations, mapped to the buttons labeled S, M, and R on the controller. This allowed users to switch between variations within each group.

Volume for each sound group was mapped to the faders on each channel. To address the unpleasantness of sharp sounds noted during field observations and the workshop, I included the option to "muffle" sounds using a low-pass filter controlled by the rotary knobs, ranging from 22,000 Hz (value 0, no filter) to 500 Hz (value 1, heavily muffled).

I did not spatialize sounds with more than two sources due to time constraints. Bags, carts, checkout, and freezer were spatialized, while music, footsteps, room tone, and voices were not. I used FMOD's spatializer plugin with adjusted attenuation curves to simulate realistic spatial sound. For example, the freezer hum had a broader spread, while checkout beeps had a narrower one.

To capture the soundscapes participants would create, I mapped the “CYCLE” button on the controller (see Figure 7) to trigger a data snapshot. A Unity script recorded the active values for all sound parameters, selected variation, volume, and filter level, and exported them to a JSON file.

3.6 User study

3.6.1 Aims and rationale

The aim of the user study was to answer the first and third research questions. All participants were non-experts, aligning with the study’s focus. I chose individual testing over focus groups to give participants time to use the tool and reflect without outside input.

Usability testing was used to examine how participants interacted with the interface (Lazar et al., 2017, p. 264). Participants were asked to create three versions of a grocery store soundscape (realistic, pleasant, and unpleasant) to see whether the tool supported expressing preferences and ideas. I also wanted to examine whether the tool encouraged reflection on everyday sound and how participants described their interaction. To gather both quantitative and qualitative data, I used a Likert questionnaire (Lazar et al., 2017, p. 120) to gain an overview of their experiences, followed by a semi-structured interview for deeper insights (Lazar et al., 2017, p. 198).

3.6.2 Procedure

Seven participants (three men, four women) aged 22 to 34 (M = 28, SD = 3.6) took part in the study. They were recruited through social media and word of mouth. All participants gave informed consent before beginning the session. Audio was recorded for transcription purposes with participant permission. All sessions were conducted in person, in a quiet room. Each session lasted around one hour and followed the same structure.

I started each session with a short introduction to the study, the tool, and the structure of the session. All sound sources were already playing when the scene loaded, with volumes set to zero, so the store began in silence. I demonstrated the tool briefly, then handed over control and encouraged the participant to explore freely.

After this initial exploration, I asked them to create three different versions of the store’s soundscape:

1. **A realistic version:** *what an average grocery store sound like*
2. **A pleasant or ideal version:** *how they would like the store to sound*
3. **An unpleasant or stressful version:** *how the store might sound at its worst*

The tasks were designed to be clear and not require additional explanation (Lazar et al., 2017, p. 286). After participants felt done with each soundscape, or after 10 minutes had passed, I saved the values. This phase typically lasted 20–30 minutes. Afterwards,

participants filled out a 5-point Likert-scale questionnaire with seven statements about their interaction with the tool, rated from "strongly disagree" to "strongly agree" (see Figure 14).

The session concluded with a semi-structured interview, which lasted around 15 minutes and was audio recorded. The questions were prepared beforehand (see Appendix A), but I also encouraged participants to speak freely and asked follow-up questions when appropriate.

3.6.3 Data Analysis

The saved values for each soundscape were collected, including volume settings, filter settings, and variation selections. Questionnaire responses were also collected. Interviews were transcribed and then sorted and analyzed using coding. The coding was guided both by the research questions and by patterns that emerged from the data itself.

4 Results

This chapter presents the outcomes of the user study. It begins with how participants adjusted volume and filter settings when creating the three soundscape versions. It then shows the specific sound variations selected for each version. Next, it presents responses from the post-use questionnaire. Lastly, it details findings from the interviews, including reflections on using the tool, strategies for shaping the soundscapes, and participants' experiences and thoughts on everyday soundscapes after using the tool.

4.1 Volume and Filter Settings

Tables 2–4 present the mean volume and low-pass filter settings for each sound group as selected by participants for each soundscape. Volume is reported as normalized gain on a scale from 0.0 to 1.0, with higher values indicating greater intensity. Low-pass filter settings are reported as cutoff frequencies in Hz, based on a linear mapping from the rotary control (0 = 22 000 Hz, no filtering; 1 = 500 Hz, heavily muffled). When volume was set to zero for a channel, the associated sound and filter had no audible effect and were excluded from the low-pass calculations. An overview of the sound channels is provided in Table 1.

4.1.1 Realistic Soundscape

In the realistic soundscape, all participants adjusted volume on every sound group except Music, which was used by three participants. Average volume levels ranged from 0.22 to 0.57. The highest means were for Freezer hum, Voices, and Room tone, while Music had the lowest. Volume settings were consistent, with standard deviations between 0.10 and 0.21, except for Music.

Table 2: Realistic soundscape. Mean volume and low pass filter cutoff frequency (Hz) with standard deviation (SD) for each channel.

Channel	Sound Group	Volume Mean	Volume SD	Cutoff Frequency Mean (Hz)	Cutoff Frequency SD (Hz)
1	Bags	0,47	0,18	20 710	19 420
2	Carts	0,49	0,18	18 345	14 045
3	Checkout	0,53	0,18	19 635	17 485
4	Freezer	0,57	0,10	16 195	14 260
5	Music	0,22	0,28	15 120	10 390

6	Footsteps	0,50	0,18	19 420	18 560
7	Room Tone	0,54	0,21	18 130	14 045
8	Voices	0,54	0,21	15 980	13 615

Mean low pass filter cutoff frequencies were high, ranging from 15 120 Hz to 20 710 Hz. The highest filtering was applied to Music, Voices, and Freezer hum. The lowest were for Bags and Footsteps. Variation was highest for Music and Voices, and lowest for Bags and Footsteps.

4.1.2 Pleasant Soundscape

In the pleasant soundscape, participants applied slightly lower and more varied volume levels, with means from 0.17 to 0.53. Voices had the highest, followed by Carts, Bags, and Footsteps. Freezer hum had the lowest. Music saw a slight increase in use compared to the realistic soundscape but remained variable. Standard deviations ranged from 0.19 to 0.28.

Table 3: Pleasant Soundscape. Mean volume and low pass filter cutoff frequency (Hz) with standard deviation (SD) for each channel.

Channel	Sound Group	Volume Mean	Volume SD	Cutoff Frequency Mean (Hz)	Cutoff Frequency SD (Hz)
1	Bags	0,39	0,19	14 905	16 625
2	Carts	0,42	0,21	17 270	15 335
3	Checkout	0,29	0,21	12 970	15 335
4	Freezer	0,17	0,22	9 530	13 400
5	Music	0,37	0,28	16 840	12 755
6	Footsteps	0,39	0,24	18 990	18 560
7	Room Tone	0,35	0,21	15 980	13 615
8	Voices	0,53	0,23	13 830	14 260

Low-pass filtering was used more frequently, with mean cutoff frequencies ranging from 9 530 Hz to 18 990 Hz. The highest filtering was applied to Freezer hum, Checkout, and

Voices. The lowest were for Footsteps and Carts. Variation was highest for Music and Freezer hum.

4.1.3 Unpleasant Soundscape

In the unpleasant soundscape, participants applied high volume levels across most groups, with means from 0.53 to 0.99. The highest were for Carts, Checkout, Freezer hum, and Voices. Room tone was lower and less dominant in this version. Variation was minimal except for Room tone, Music, and Footsteps.

Table 4: Unpleasant Soundscape. Mean volume and low pass filter cutoff frequency (Hz) with standard deviation (SD) for each channel.

Channel	Sound Group	Volume Mean	Volume SD	Cutoff Frequency Mean (Hz)	Cutoff Frequency SD (Hz)
1	Bags	0,92	0,14	20 495	18 345
2	Carts	0,99	0,03	19 635	17 915
3	Checkout	0,98	0,05	18 130	15 335
4	Freezer	0,97	0,05	17 915	13 830
5	Music	0,77	0,37	20 280	18 130
6	Footsteps	0,86	0,22	20 925	19 205
7	Room Tone	0,53	0,46	18 345	17 055
8	Voices	0,97	0,05	20 710	19 850

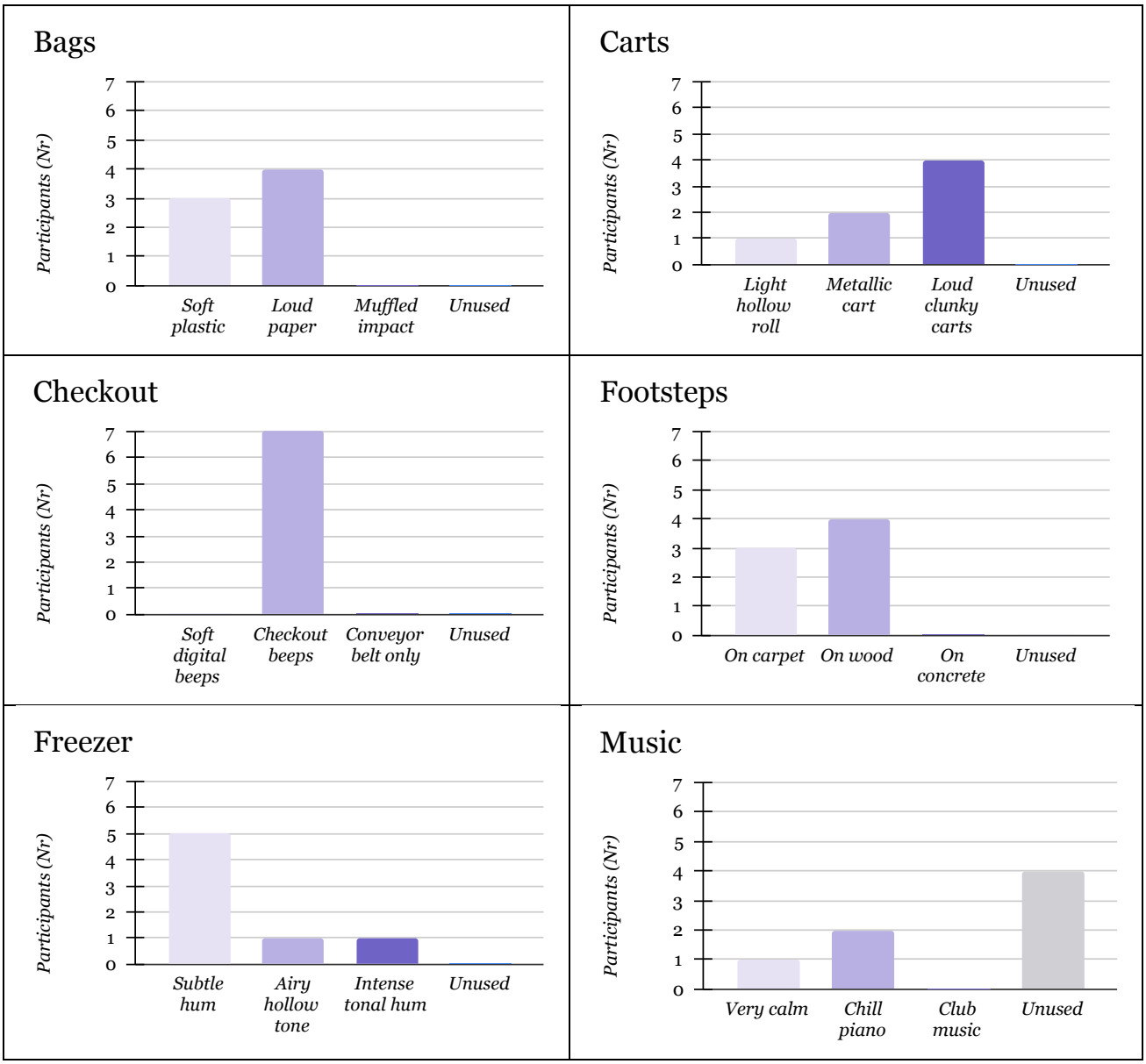
Low-pass filtering was rarely used, with mean cutoff frequencies ranging from 17 915 Hz to 20 925 Hz. The highest variation was observed for Freezer hum and Checkout, while other groups, including Music, showed lower variation.

4.2 Choices of Sound Variations

Figures 11–13 present the selected sound variations per sound group (channel) for each soundscape version. Instances where participants set the volume to zero are shown as Unused in the graphs. An overview of the available sound variations per channel is provided in Table 1. Some variation names in the figures have been slightly adjusted to fit the diagram format.

4.2.1 Realistic Soundscape

Figure 11 shows the distribution of selected variations across channels. Clear convergence was observed for Checkout, where all participants selected Checkout beeps, and for Freezer, where most participants selected Subtle hum. A similar majority pattern was present for Room Tone and Voices, where five participants selected Small indoor area and Airport gate, respectively. In contrast, Bags, Carts, and Footsteps showed more variation, with selections spread across different variations. Music was used by fewer than half of the participants, with one participant selecting Meditative and two selecting Chill piano.



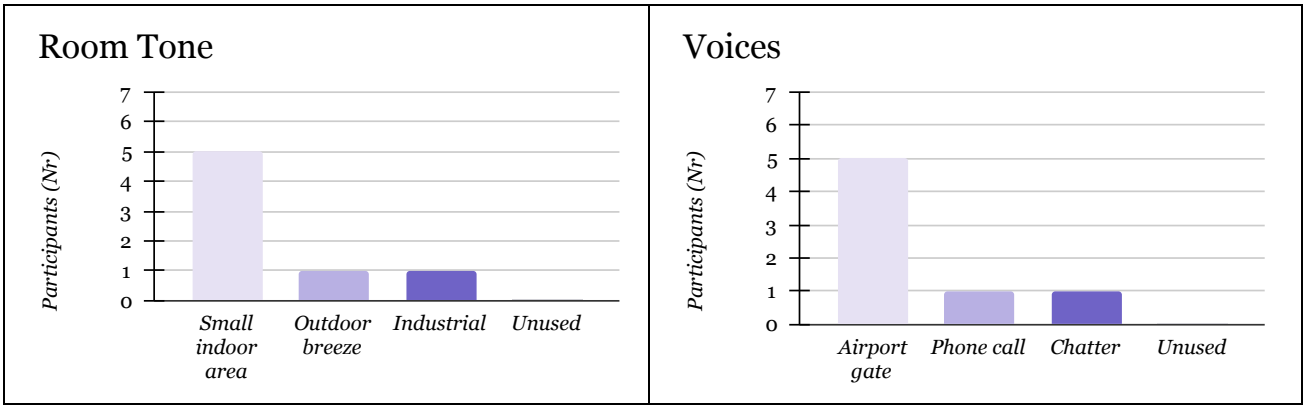
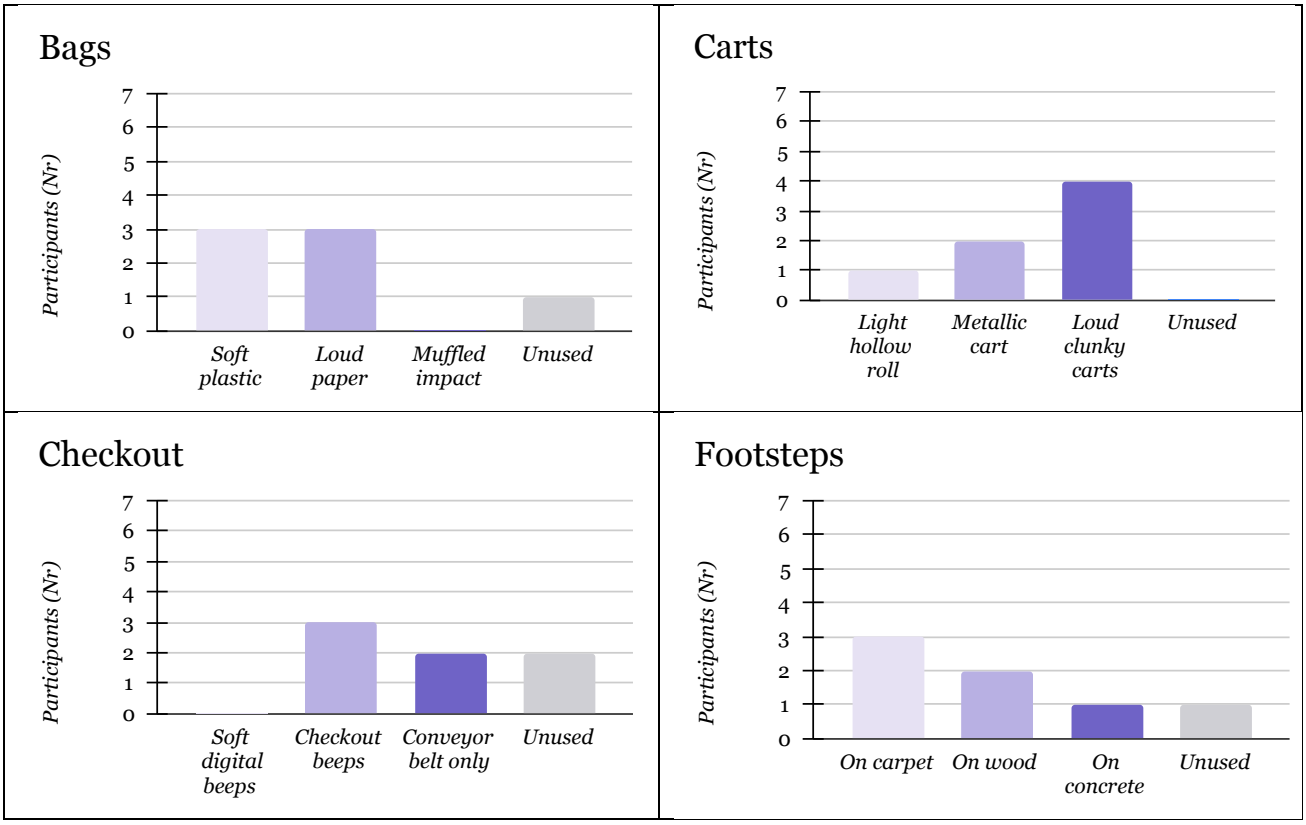


Figure 11: Variation Selections per Channel for the Realistic Soundscapes

4.2.2 Pleasant Soundscape

Figure X shows the distribution of selected variations across channels for the pleasant condition. Compared to the realistic condition, fewer channels showed clear convergence. Voices were included by all participants, with selections divided between Airport gate and Chatter. Music was included by most participants, with most selections using Chill piano. Freezer was often not included. The remaining channels showed more variation, with choices spread across different versions.



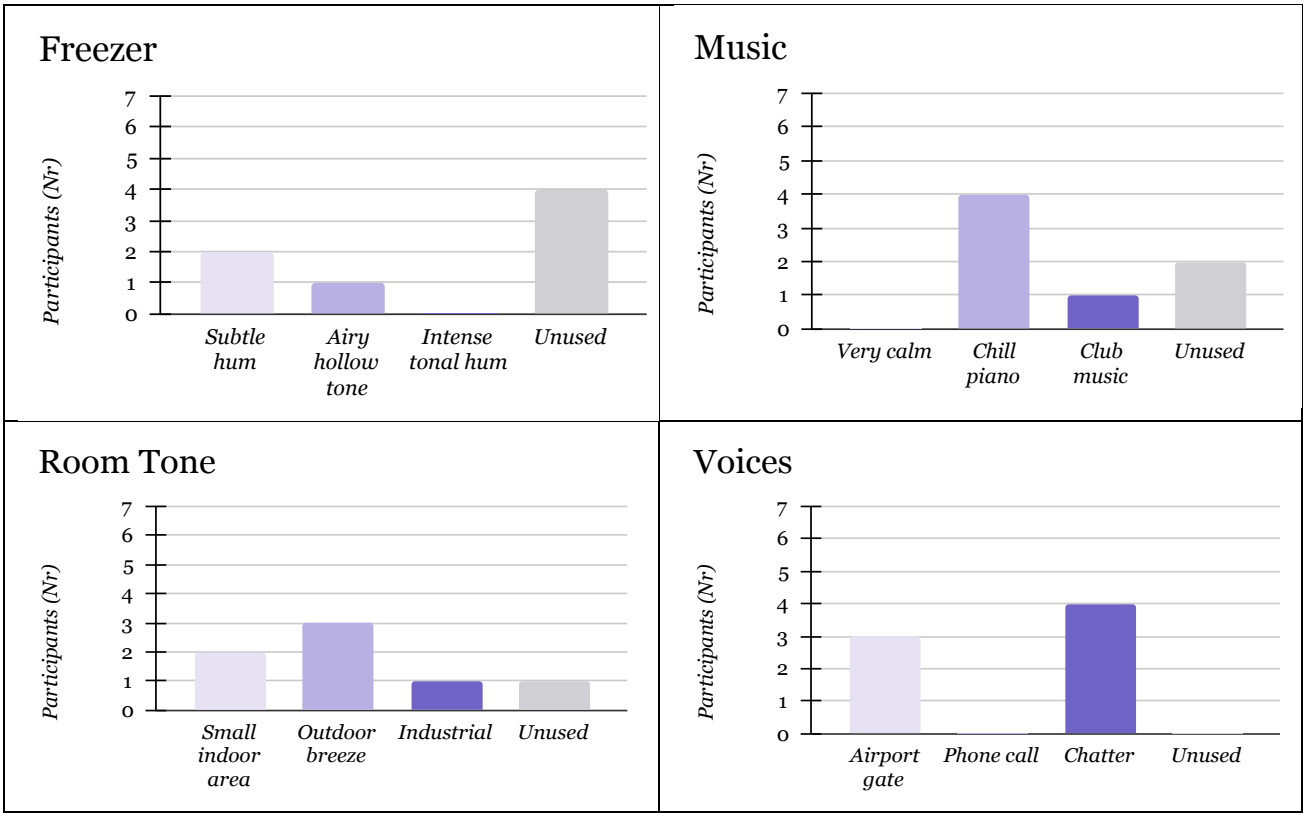
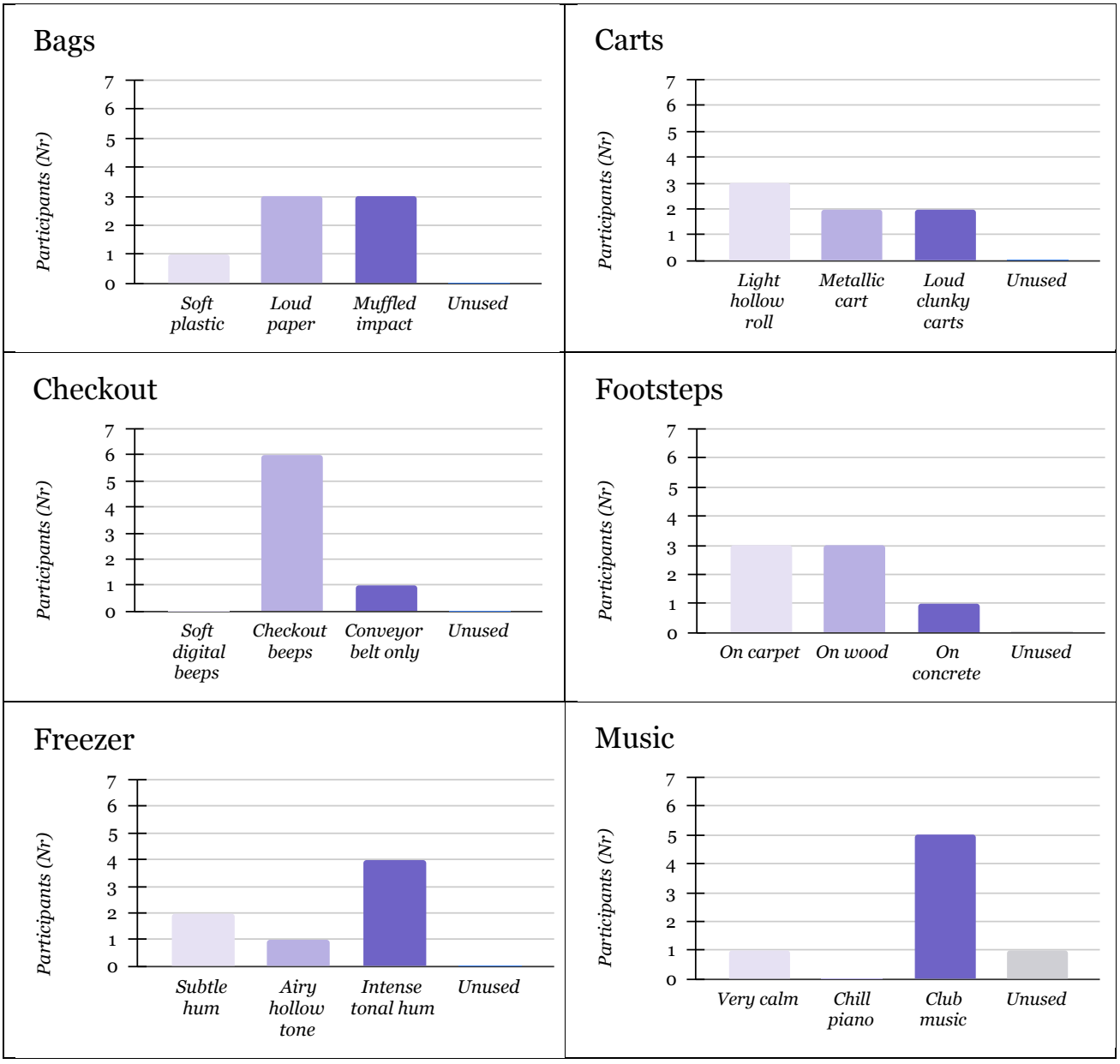


Figure 12: Variation Selections per Channel for the Pleasant Soundscapes

4.2.3 Unpleasant Soundscape

Figure X shows the distribution of selected variations across channels for the unpleasant soundscapes. A majority pattern was observed for Voices, where six participants selected Phone call, and for Music, where five participants selected Rhythmic beat. Checkout showed a similar majority pattern, with six participants selecting Checkout beeps. Freezer was most often set to Intense tonal hum, while other freezer variations were used by fewer participants. In contrast, Bags, Carts, Footsteps, and Room Tone showed more variation,

with choices spread across different versions and, for Room Tone, two participants did not include the sound.



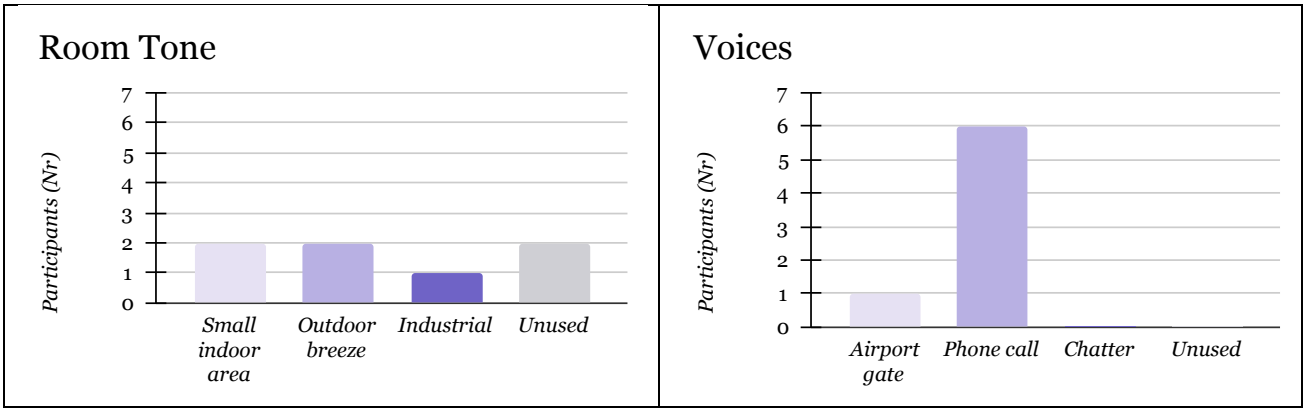


Figure 13: Variation Selections per sound group for the Unpleasant Soundscapes

4.3 Likert Questionnaire

Figure 14 shows participant responses to the post-use questionnaire. The questionnaire consisted of Likert-scale statements with five response options, ranging from Strongly disagree to Strongly agree. Responses were overall positive. Participants largely agreed with the positively framed statements about shaping the soundscape, creative control, enjoyment of the process, reflection on sound, and the sense that their choices were intentional and meaningful. For the negatively framed statements, most participants disagreed, suggesting that the tool was not experienced as difficult to use or limiting.

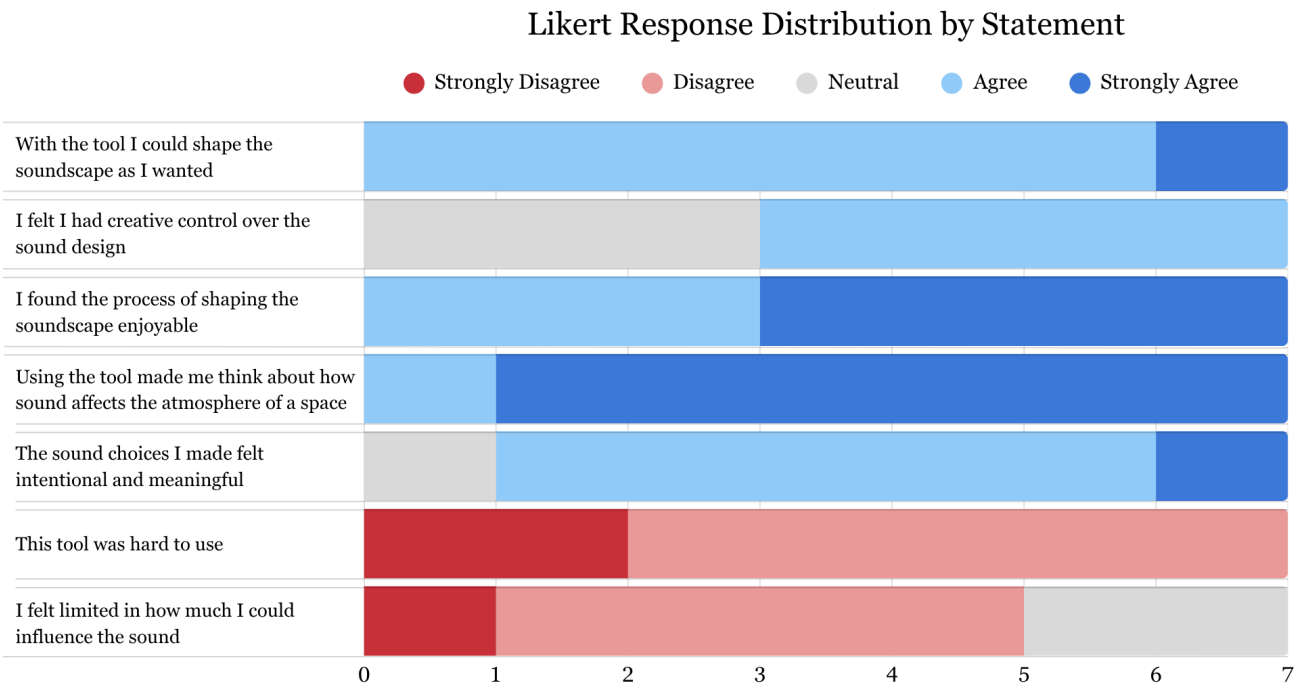


Figure 14: Participant Responses to the Post-Use Questionnaire on Soundscape Creation with the Tool

4.4 Interview Results

This section presents results from the interviews. The findings are organized thematically, reflecting participants' experiences, strategies, and reflections after using the tool.

4.4.1 Reflections on Using the Tool to Create Soundscapes

All participants reported being able to manipulate the soundscape in *meaningful* and *intentional* ways, describing the interaction as *intuitive* and *accessible*.

Several highlighted the value of having multiple adjustment parameters, noting the interface supported a wide range of combinations even if they did not explore every possibility. P1 described successfully creating a soundscape that felt familiar and characteristic of a store environment. They found the process intuitive and aligned with their expectations. They also noted that the interplay between volume, variation, and filtering provided a strong sense of control. P2 felt the available sounds were accurate and sufficient, and P4 described the outcome as realistic.

Participants also said they could recreate soundscapes matching the prompts and found the tool easy to navigate. P4 stated it was easy to use with no confusing features, and P6 noted that the controls felt natural and the design supported moving through the steps in order.

However, participants also offered feedback on areas for improvement. P3 noted challenges in representing how the soundscape might shift with spatial position and chose to focus on one location. P5 and P6 mentioned missing elements such as store staff and checkout interactions, with P5 suggesting snippets like "*Do you want a receipt?*" and interactive sounds like fridge doors opening and closing to increase realism.

4.4.2 Strategies When Using the Tool

Participants described volume as their primary method for shaping the soundscape. Beyond that, their strategies varied. P5 began by adjusting ambient sounds, P6 worked methodically from left to right on the controller, and P7 focused on selecting appropriate sound variations rather than adjusting every parameter.

The low-pass filter was described as especially useful for creating the pleasant version, helping reduce the sharpness of specific sounds. P6 noted using the filter extensively, and P3 associated it with a "*muffled*" effect that contributed to a more comfortable environment. P5 used the filter to soften sharp or high-frequency sounds but avoided applying it to already pleasant sounds to maintain a balanced mix.

Participants' approaches to constructing the three versions also revealed how they conceptualized and differentiated them. They frequently compared the versions and adjusted them in relation to each other. For example, sounds identified as disturbing in the

realistic version were often amplified in the unpleasant version, while overwhelming elements were softened or filtered out in the preferred version.

4.4.3 Unpleasant and Pleasant Soundscapes

The *unpleasant version* was typically created by *exaggerating elements participants found irritating or overwhelming*. Many tried to reproduce stressful conditions, such as shopping during peak hours, by increasing volume, layering sharp sounds, or removing filtering. While exaggerated, these versions were often described as grounded in real experiences. For the *pleasant version*, participants *emphasized soft textures, overlapping voices, and reduced sharpness* to create spaces that felt calm, private, and comfortable. Some likened these versions to cafés or quiet public spaces.

Participants also reflected on individual sounds. Checkout beeps were mentioned in nearly all interviews as particularly disturbing. P2 said beeping sounds are generally triggering, P3 described them as the worst sound, and P6 said they were awful. Loud bag sounds were described as sharp and irritating, while softer versions were seen as more neutral. Metal carts and clunky rolling sounds were often described as unpleasant, while softer or "*plastic-like*" cart sounds were preferred. The freezer hum was considered overwhelming or harsh when loud or high-pitched, but when softer and lower in frequency, some found it neutral or calming.

Voices elicited different reactions depending on the version. A man talking on speakerphone was frequently described as disruptive or cognitively demanding. P1 said it felt intrusive, P4 said it was like someone following them while talking loudly, and P6 noted that hearing only half of the conversation drew attention. In contrast, overlapping voices or background murmur were often described as neutral or pleasant. P5 selected a version resembling café chatter, and P6 preferred voices that were not fully intelligible.

Music preferences varied, but most participants found the electronic track unpleasant in a store environment. Even those who liked the genre felt it did not fit the context. For the unpleasant version, P6 selected the most "*overly calm*" music, describing it as "*dystopian and corporate*" when combined with a chaotic soundscape, as if the store believed calm music alone would make the environment pleasant.

The room tone was never described as unpleasant and was seen by most participants as important for avoiding a sense of emptiness. P6 and P7 noted that they did not include it in the unpleasant version, as it helped mask more disturbing sounds.

4.4.4 Personal Influences on the Soundscape Choices

Participants described how *personal memories*, em*otional associations*, and *prior experiences* shaped their approach to constructing soundscapes.

P1 and P4 drew on familiar store environments, tailoring the soundscapes to match these memories. P7 mentioned that working in a grocery store likely shaped their aversion to beeps. P6 said they envisioned themselves in a good or bad mood when creating the pleasant and unpleasant versions.

P3 noted they liked human voices and associated them with comfort while finding mechanical sounds harder to tolerate. P5 chose paper bags because they are better for the environment and associated the rolling of metal carts with horror movies. They also described freezer hums as either nostalgic or discomforting, depending on the context.

Several participants mentioned typically wearing headphones while shopping. P2 said they kept this in mind when creating the realistic version. For the pleasant version, P2 considered the muffled effect of noise-cancelling headphones, selecting sounds that "*were not too disturbing, but still with some sounds so you can orient yourself*".

4.4.5 Reflections on Soundscape Awareness

The tool prompted participants to reflect on soundscapes in ways they had not done before. Most said the tasks made them more *aware of everyday soundscapes* and *attuned to details they might otherwise overlook*.

P1 noted that combining and layering sounds helped them understand how soundscapes are composed, while P3 felt that the interaction between sounds could be more stressful than hearing them individually. P6 reflected on how context shapes the experience of sound, and most participants said the process would likely influence how they think about supermarket soundscapes in the future. P5 noted the challenge of describing sound, observing that their descriptions often relied on associations rather than technical terms.

Several participants said that creating the realistic versions revealed how stressful or overstimulating grocery store soundscapes can be. P3 and P6, for example, reflected that the realistic and unpleasant versions were not so different in practice.

5 Discussion

In this section I discuss the findings in relation to the research questions presented in the introduction. The discussion is structured around these questions, drawing on results from the user study, interviews, and the created soundscapes. It also reflects on limitations and identifies directions for future work.

RQ1: *Can an interactive tool support non-experts in expressing their soundscape preferences and reflecting on their everyday sound environments, and if so, how?*

RQ2: *What considerations and methods inform the design process of such a tool?*

RQ3: *What insights can be gained from the soundscapes created by participants using the tool?*

5.1 Using an Interactive Tool to Reflect on and Shape Soundscape Preferences

Previous research highlights that non-experts often lack both the vocabulary and the habit of reflecting on soundscapes (Blesser & Salter, 2007). The tool in this study addressed this by offering direct interaction with parameters such as volume, filtering, and variation selection. This enabled participants to shape soundscapes in ways they described as intentional and reflective. It also provided a practical way to capture subjective perceptions of pleasantness, as emphasized by Filipan et al. (2017).

The findings suggest that even relatively simple controls can make soundscape reflection more accessible. All participants were able to create soundscape versions that matched their intentions, often describing the experience as intuitive and engaging. These results support those of Tarlao et al. (2023), who showed that interactive soundscape tools can encourage reflection among professionals. However, this study extends that insight by showing how interactive tools can support reflection and expression among non-experts working independently, without direct facilitation.

Beyond confirming earlier findings, this study contributes a practical method for engaging non-experts in exploring and shaping the sounds around them. The results highlight how interactive tools can lower barriers for critical engagement with soundscapes by prompting participants to attend to individual sounds and consider how auditory environments influence mood, social interaction, and the atmosphere of a place. These aspects are often overlooked in conventional, measurement-based approaches (Steele et al., 2023). However, the interviews showed that participants tended to describe their choices through personal associations rather than using shared or technical terms. This suggests that while the tool supported reflection, it did not necessarily promote a common vocabulary for discussing sound.

Overall, the findings indicate that interactive, embodied tools can help non-experts express preferences, reflect on their sound environments, and become more aware of everyday soundscapes. This points to a broader opportunity for participatory soundscape work, where people directly affected by acoustic environments, such as employees, residents, or community members, are included in design processes. Rather than replacing expert assessments, tools of this kind may serve as a bridge between subjective experience and professional practice. This highlights the potential of interactive tools to support more inclusive and reflective approaches to the design and management of everyday soundscapes.

5.2 Considerations and Methods in Design Process

The design process in this study aligns with the view that soundscapes are best understood in everyday environments rather than in laboratory settings (Gaver, 1993a). It follows the ISO (2014) definition of soundscapes, which emphasizes perception in context. By moving from field observations to sound recording and editing, a co-design workshop, prototype development, and a user study, each stage built on and clarified the previous one. This layered and iterative approach allowed the study to stay grounded in everyday listening while remaining responsive to participant perspectives. It also ensured that methods complemented and informed each other (Bryman, 2006; Greene et al., 1989; Niglas, 2004).

An important insight from this process is that combining field listening with participatory methods can help translate subjective experiences of soundscapes into practical design decisions. The workshop showed that participants had strong reactions to specific sounds and expressed a wish to control which sounds were included, how prominent they were, and how they blended with other sounds in the environment. This informed the inclusion of adjustable controls for volume, sound variations, and filtering in the prototype. The decision to use a physical MIDI controller supported embodied interaction and reflected participants' suggestions for intuitive, hands-on control. These choices respond to a field where technical measurements often take precedence over how people actually perceive and engage with sound in everyday settings. This suggests that designing for soundscape reflection requires both attention to how people experience sound in context and careful consideration of how interaction methods support agency and exploration. It also highlights that participatory methods are not only a source of input but an important part of shaping tools that feel accessible and meaningful to non-experts.

Beyond shaping the tool itself, this process illustrates how layered and iterative methods can translate subjective reflections on soundscapes into practical design choices. Grounding the work in everyday listening and participant input ensured that the tool remained accessible and meaningful for non-experts. This complements the findings discussed in relation to RQ1 by showing that the ability of the tool to support non-experts in expressing

and reflecting on soundscape preferences was made possible by a design process that combined participatory, mixed-methods, and interdisciplinary approaches.

5.3 Insights from Participants' Created Soundscapes

The created soundscapes reveal that perceptions of pleasantness and disturbance were shaped by how sounds were combined, adjusted, and experienced in relation to each other. While loudness was consistently used to express stress and unpleasantness, reflecting regulatory framings of noise (OSHA, 2025; The Swedish Work Environment Authority, 2023), the results also show that qualities such as sharpness and how sounds captured attention influenced comfort. This supports Gyllensten et al. (2023), who note that "meaningful but disturbing sounds" can feel stressful even at lower volumes, as seen in the consistent highlighting of sharp checkout beeps or the intrusive phone call.

The results also showed that the pleasant soundscapes could be understood more from a perspective of ecotonality (McCartney, 2016) rather than following a "hi-fi" pattern focused on clear separation of sound sources (Truax & Barrett, 2011). In several cases, participants layered sounds to soften sharp edges and create a lively yet comfortable atmosphere, even when this meant that individual sound sources became less distinguishable. Rather than aiming for clarity, participants valued a blended richness when it contributed to a sense of comfort within the environment. This echoes the findings of Marry and Defrance (2013), who showed that soundscapes can be both lively and pleasant.

The results further illustrate that perceptions of sounds were shaped by what participants expected to hear in a grocery environment. For example, music that might be enjoyable elsewhere was described as out of place or "dystopian" in the store context, while the hum of freezers was adjusted rather than removed, as it was perceived as part of the environment. This reflects the role of contextual meaning in how soundscapes are experienced (Southworth, 1969; Bruce & Davies, 2014).

One unexpected finding was that in the pleasant version, no participant chose the softer checkout beep. Despite consistently describing the standard beep as unpleasant in both the workshop and the user study, participants continued to include it. Given these negative descriptions, one might expect them to avoid it, especially since Arfvidsson et al. (2021) found that softer, more blended alarm sounds in stores can still be effective. One possible reason is that participants expected the standard beep in a grocery store, which could relate to hi-fi soundscapes as discussed by Truax and Barrett (2011). Another possibility is that the softer beep felt unnatural or did not blend well, making the familiar sharp beep preferable despite its unpleasantness. This resonates with Kontukoski (2018), who suggests that sounds that blend into the environment are often experienced as more pleasant than ones that stand out.

Personal associations also shaped how participants constructed their soundscapes, reflecting how memories and prior experiences informed their preferences. Some aimed to replicate pleasant store environments they knew, amplified human voices because they found them comforting, or chose the paper bag sound due to its association with environmental values. This aligns with Blesser and Salter's (2007) discussion of how personal and social context influences how people react to sounds.

The results also showed that headphone use shaped how some participants imagined and reflected on soundscapes. One participant, for example, designed the pleasant version to resemble how a store sounds when using noise-cancelling headphones. This connects to Barns (2019), who describes the use of headphones in public spaces as creating "private sound environments". More broadly, it highlights how experiences and reflections on soundscapes are shaped by contemporary listening habits and by people's roles and activities within a space.

Taken together, these findings suggest that by creating complete soundscapes in the grocery store environment, participants could explore how sounds interact to shape the overall atmosphere. It is likely that asking participants to rate sounds individually, or to create soundscapes for a different environment, would not have revealed the same preferences or insights. The created soundscapes offer a way to examine how people understand and shape everyday acoustic environments, showing how perceptions of what feels pleasant or stressful develop through the practical process of adjusting and combining sounds within a specific context.

5.4 Limitations

While the study demonstrates how an interactive tool can support non-experts in shaping and reflecting on soundscapes, several limitations must be acknowledged.

First, the project was conducted within the timeframe of a master's thesis, which limited both the technical refinement of the prototype and the range of features that could be explored. Time constraints also affected the depth of the participatory process, particularly in the workshop phase. Only one workshop was conducted with a small group of participants, reducing the diversity of perspectives that could have informed both the design choices and the interpretation of findings.

Second, the participant group in the user study was small and relatively homogeneous in terms of age and background. This limits the transferability of the findings, especially regarding how different user groups such as children, older adults, or people with varied cultural experiences might interact with the tool or interpret soundscapes.

Third, several participants noted the absence of common store elements in the prototype, such as staff voices, checkout interactions, and interactive sounds. These missing elements

may have influenced how participants constructed their soundscapes and limited the range of expressions captured. As discussed during development, sourcing realistic vocal recordings proved challenging, and the inclusion of interactive sounds was beyond the technical scope of the project. With more time, it would have been valuable to include a broader set of sound sources, expand spatialization, and allow participants to interact with dynamic sounds rather than premade loops.

A further limitation concerns the taxonomy used for selecting sounds. Although it was informed by the Urban Sound Taxonomy (Salamon, Jacoby, & Bello, 2014), the categories were shaped by my own field observations and reflections. This means that the selection of sounds reflected a subjective interpretation of the grocery store environment, based on a limited set of observations, rather than a systematic survey. This may have influenced both the design of the prototype, and the types of soundscape compositions participants were able to create.

These limitations highlight that the study's insights are closely tied to its context, participant group, and prototype design. They also point to areas where future studies could extend the investigation, including broader participant groups, different environments, and expanded tool functionality. The findings reflect how the tool supported reflection and expression within the scope of this project, but further work is needed to explore how participatory soundscape tools perform in other contexts and with different user groups.

5.5 Future Work

This study suggests several directions for future research. One is to examine how interactive tools for soundscape reflection and shaping can be applied in environments with different acoustic and social characteristics, such as public transport, outdoor spaces, or workplaces. Expanding the range of contexts would make it possible to explore how environmental factors influence both user interaction and perceptions of soundscape quality.

A second area concerns participant diversity. Involving a wider range of users, including participants of different ages, cultural backgrounds, and levels of technological experience, could offer insights into how accessible and meaningful the tool is for different groups. Such studies may also reveal whether the tool supports reflection and soundscape expression across varying sensory, cognitive, or physical abilities.

A next step in this work could be to evaluate the soundscapes created using the tool. In this study, participants constructed soundscapes based on their own preferences, but the resulting soundscapes were not evaluated in terms of perceived pleasantness, realism, or overall quality. Future studies could involve having participants, or separate listener groups, experience and assess these soundscapes. This would make it possible to examine whether soundscapes that participants create as pleasant, realistic, or unpleasant are also

experienced that way by listeners, and to identify which sounds contribute to these qualities.

Another direction is to investigate how the tool can function in participatory soundscape design processes. Future work could explore whether using the tool helps participants express preferences in collaborative design settings and whether this leads to sound environments that more clearly reflect user intentions.

Finally, further research could examine whether interactive tools can support the formulation of soundscape requirements that extend beyond noise level measurements. This may be particularly relevant in environments such as workplaces, schools, or healthcare settings, where sound regulations often focus on exposure limits but do not account for how sound affects concentration, communication, or well-being.

6 Conclusion

This thesis explored whether an interactive tool could support non-experts in expressing soundscape preferences and reflecting on their everyday sound environments. The research combined fieldwork, participatory design, prototype development, and a user study, showing that direct interaction enabled participants to express preferences and reflect on soundscapes in meaningful ways.

The study also demonstrated how participatory methods informed the tool's design and how the created soundscapes revealed the role of context, expectation, and personal associations in shaping how participants perceived and adjusted everyday sound environments. Participants emphasized how sounds interacted within a setting, rather than evaluating them in isolation, highlighting the importance of situational context and individual experience.

Together, these findings show that interactive tools, when grounded in participatory and iterative methods, can support reflection, expression of preferences, and awareness of everyday soundscapes. They also suggest a pathway for including non-expert perspectives in soundscape design and research, by making subjective experiences accessible and actionable within design processes.

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Appendix A

Interview Questions:

1. Can you describe the soundscape you created? What kind of atmosphere were you going for?
2. What did you focus on when adjusting the sounds?
3. Were there any sounds you found particularly important, disturbing, or surprising?
4. How did the tool help (or not help) you achieve the sound you wanted?
5. Did any part of the sound design process feel particularly intuitive or difficult?
6. Was there anything that influenced your choices, for example, specific memories, places, or emotional states?
7. Did this experience in any way make you more aware of sound in everyday spaces? If yes, then how?

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